

```
75. sp500['Returns'] = sp500['Adj Close'].pct_change()
76. merged = pd.DataFrame({
77.     'Stock': data['Returns'],
78.     'Market': sp500['Returns']
79. }).dropna()
80. correlation = merged.corr().iloc[0, 1]
81. print(f"\nCorrelation with S&P 500: {correlation:.4f}")
82.
83. # Calculate beta
84. beta = merged.cov().iloc[0, 1] / merged['Market'].var()
85. print(f"Beta: {beta:.4f}")
```